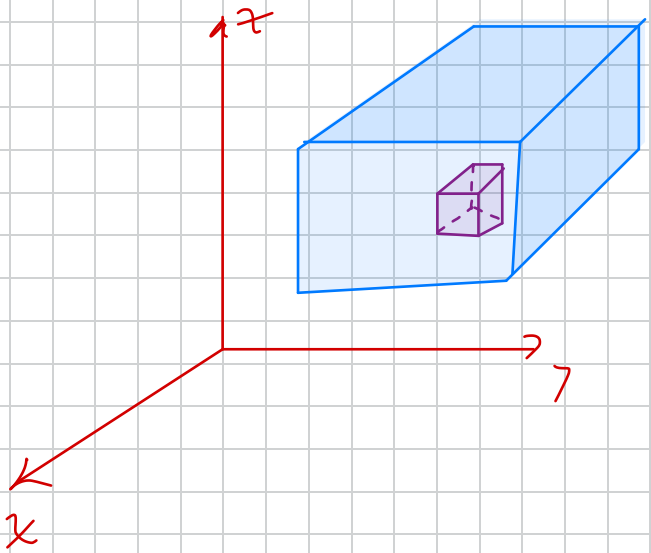
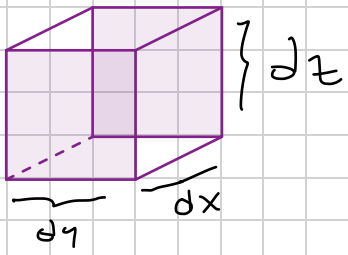
Integración triple:



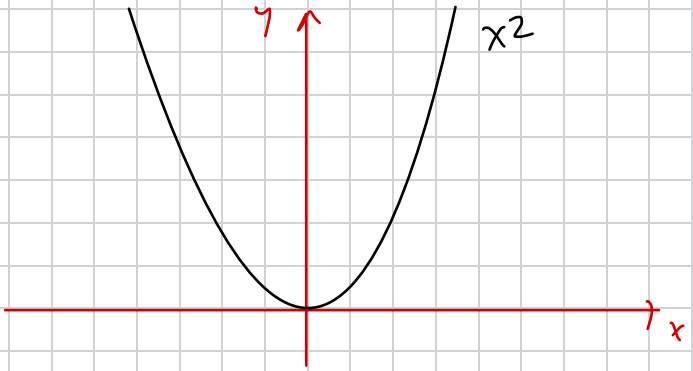


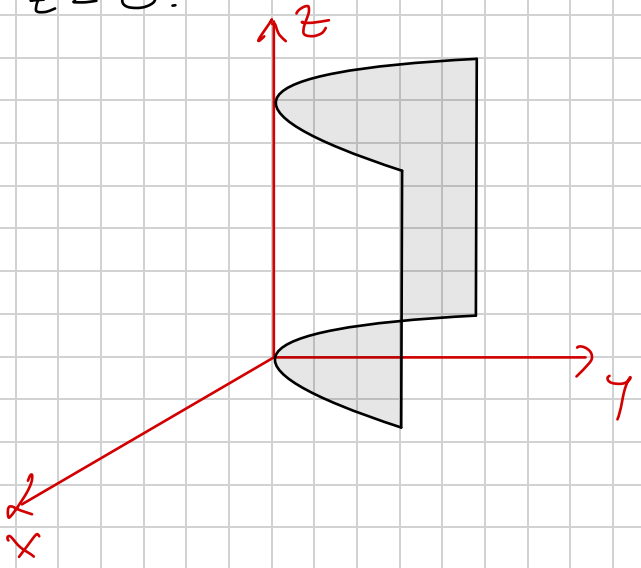
$$dV = dz\,dy\,dx$$

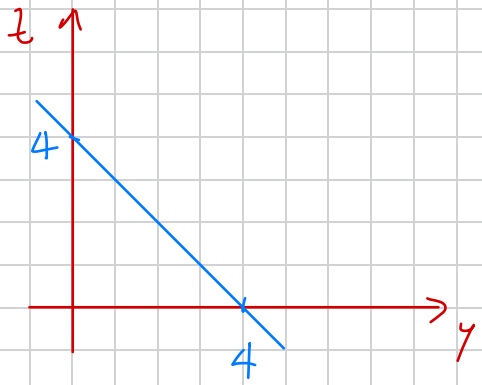
$$V = \int_a^b \int_{h_1(x)}^{h_2(x)} \int_{K_1(x,y)}^{K_2(x,y)} dz\,dy\,dx$$

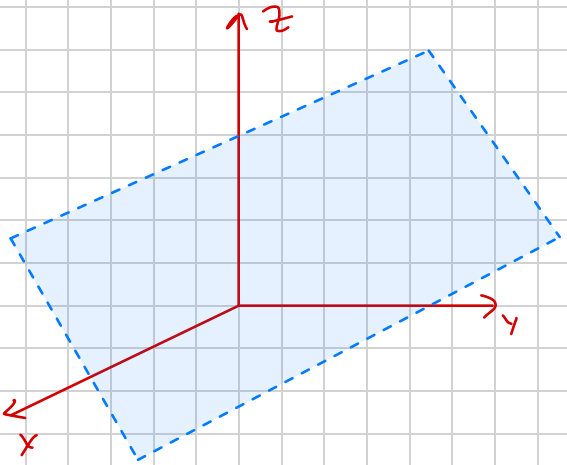
$$\iiint_Q f(x,y,z)\,dV = \int_a^b \int_{h_1(x)}^{h_2(x)} \int_{K_1(x,y)}^{K_2(x,y)} f(x,y,z)\,dz\,dy\,dx$$

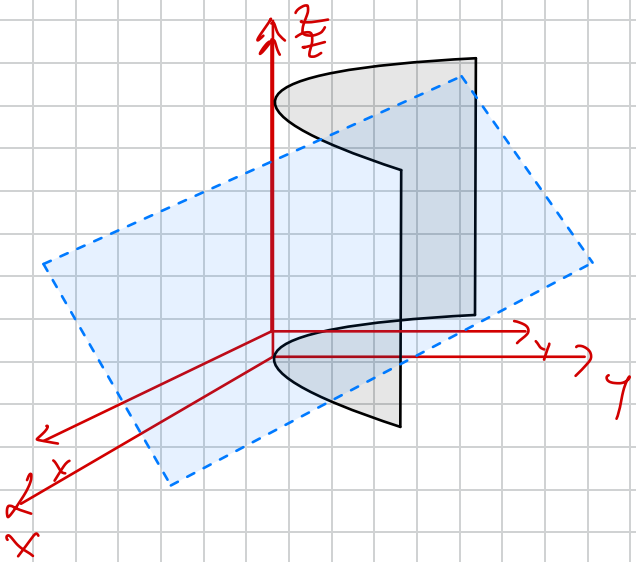
Ejemplo: Calcular el volumen de sólido acotado por la hoja parabólica $y = x^2$ y los planos $y + z = 4$ y $z = 0$.

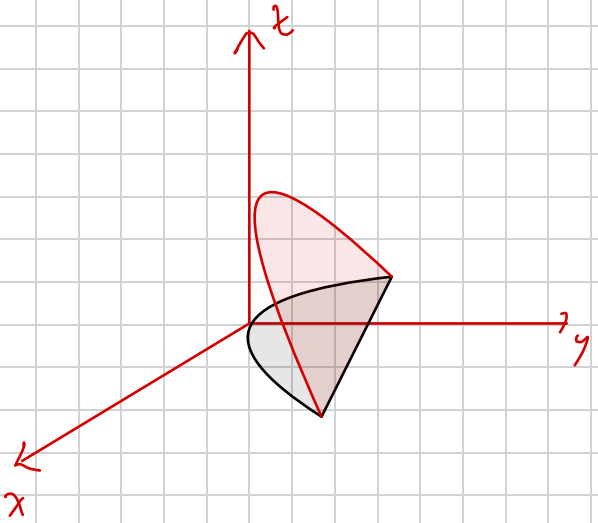


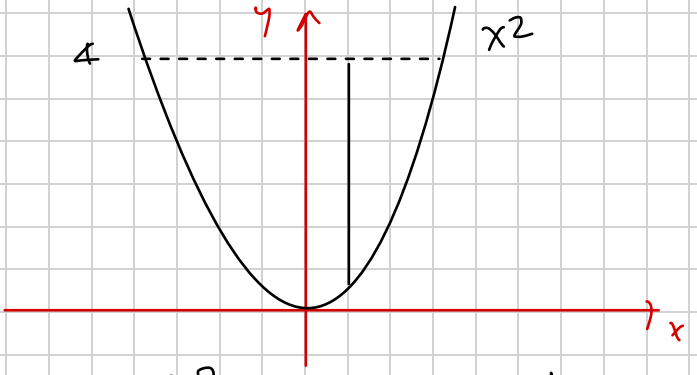












$$V = \int_{-2}^{2}\int_{x^2}^{4}\int_{0}^{4-y} dz\,dy\,dx$$

$$V = \int_{-2}^{2}\int_{x^2}^{4}(4-y)\,dy\,dx$$

$$V = \int_{-2}^{2}\left(4y - \frac{y^2}{2}\right)\Big|_{x^2}^{4} dx$$

$$16 - 8 - 4x^2 + \frac{x^4}{2}$$

$$V = \int_{-2}^{2}\left(8 - 4x^2 + \frac{x^4}{2}\right)dx$$

$$V = \left(8x - \frac{4}{3}x^3 + \frac{x^5}{10}\right)\Big|_{-2}^{2}$$

$$V = (8)(2) - \frac{4}{3}(2)^3 + \frac{(2)^5}{10} - \left(8(-2) - \frac{4}{3}(-2)^3 + \frac{(-2)^5}{10} \right)$$

$$V = 16 - \frac{(4)(8)}{3} + \frac{32}{10} + 16 - \frac{(4)(8)}{3} + \frac{32}{10}$$

$$V = \frac{256}{15} \; ud^3.$$

Ejercicio propuesto:

$$V = \int\int\int dz\, dx\, dy$$